

Human Development Index (HDI) Predictor

Data Flow Diagram

Project: Human Development Index (HDI) Predictor

Phase: Requirement Analysis

1. High-Level Data Flow

User input (form) → Flask /predict route → Country name encoded via the saved LabelEncoder → numeric feature vector assembled into a Pandas DataFrame → fed into the saved LinearRegression model (HDI.pkl) → predicted HDI score clamped to [0, 1] → mapped to a development tier → rendered back to the user on resultnew.html.

2. Training-Time Data Flow

HDI.csv → EDA (strip plot, correlation heatmap) → null values filled with column means → Country label-encoded → 90/10 train/test split → LinearRegression.fit() → evaluation (R^2 , MAE, RMSE) → model and encoder serialized with Pickle → HDI.pkl / country_encoder.pkl saved into the Flask/ folder for the web app to load.

3. Level-0 DFD (textual)

- External Entity: User (browser)
- Process 1: Collect indicator input (Prediction form)
- Process 2: Encode + predict (Flask /predict route + model)
- Data Store: HDI.pkl (trained model), country_encoder.pkl (label encoder)
- Output: HDI score + development category (Result page)